

Radiation Loss Induced by Sidewall Roughness in Planar Dielectric Waveguides

Bikesh Shrestha¹ and Ata Zadehgoi²

University of Idaho, Moscow, 83844, USA

¹ bikeshshrestha@uidaho.edu ² azadehgoi@uidaho.edu

Abstract—Planar dielectric waveguides are key to silicon photonics but suffer from scattering losses due to sidewall roughness. This work revisits the classical analysis of Lacey and Payne for TE modes, providing a complete derivation of radiation loss using Green's functions, Fourier analysis, and the Wiener–Khinchin theorem. Exponential and Gaussian autocorrelation models are examined, yielding closed-form expressions that link roughness statistics to attenuation. Results show Gaussian models predict higher peak loss and faster decay, while exponential models offer simpler, experimentally validated approximations. These findings provide scaling laws and practical design guidance for minimizing scattering and enabling ultralow-loss integrated waveguides.

I. INTRODUCTION

Planar dielectric waveguides are the fundamental building blocks for silicon photonic and optical integrated circuits [1]. But the primary challenge to achieve high performance is to avoid scattering loss, which arises from random surface roughness. This is still a primary factor, more significant than material absorption or bending loss [2]. Although roughness-induced loss has been a topic of interest and has been researched, like the work of Lacey and Payne [1], that paper has several derivation steps left out.

This work develops a complete analytical framework for radiation loss in planar dielectric waveguides with sidewall roughness. Using Green's function methods and Fourier analysis, we derive explicit formulas that relate roughness statistics to loss. Both exponential and Gaussian autocorrelation models are examined, with results showing distinct predictive trends. The exponential model, in particular, offers compact closed-form expressions consistent with experiments, making it a practical tool for guiding low-loss photonic waveguide design.

II. METHODOLOGY/APPROACH

A. Theory and Derivation

As given in [1], the refractive index distribution is,

$$n^2 = n_2^2 + (n_1^2 - n_2^2)U[d + f(z) - |y|] \quad (1)$$

where $f(z)$ is the roughness function and U is the unit step function.

Following the approach in [1], using that in the wave equation $\nabla^2 E + k_0^2 n^2 E = 0$, we get

$$\begin{aligned} \nabla^2 E_x(y, z) + k_0^2 n_2^2 E_x(y, z) \\ = k_0^2 (n_2^2 - n_1^2) U[d + f(z) - |y|] E_x(y, z). \end{aligned} \quad (2)$$

We demonstrate how the autocorrelation function $R(u)$ can be used to evaluate the roughness loss function $f(z)$ later in this paper.

B. Fourier Transform of the Green's Function

We have the Green's Function [1]

$$\nabla^2 G(y, y', z, z') + n_2^2 k_0^2 G(y, y', z, z') = \delta(z - z')\delta(y - y') \quad (3)$$

Taking the Fourier Transform of the above function and then integrating by parts twice, we get the equation as [1]

$$\tilde{G}(k_y, k_z, y', z') = \frac{e^{-(k_y y' + k_z z')}}{k_0^2 n_2^2 - k_y^2 - k_z^2} \quad (4)$$

C. Derivation of the Roughness Power Spectral Density

As we know, $f(z)$ is the roughness function representing surface roughness or imperfections and is random in nature. Instead, we calculate the power spectral density (PSD) of $f(z)$ to describe the average roughness pattern so that we can predict how much light wave scatters in total.

The Fourier transform of $f(z)$ is given by [3]

$$F(k) = \int_{-\infty}^{+\infty} f(z) e^{-ikz} dz. \quad (5)$$

For practical purposes, the limits are chosen from $-L$ to L to represent a finite part of the roughness. Thus the equation becomes [1],

$$F_L(k) = \int_{-L}^L f(z) e^{-ikz} dz. \quad (6)$$

If we square the magnitude of $F_L(k)$, we obtain the total spectral energy of the roughness over the interval $[-L, L]$. However, in order to obtain the power spectral density, we divide the spectral energy with $2L$ [1]

$$W(k) = \lim_{L \rightarrow \infty} \frac{1}{2L} \left| \int_{-L}^L f(z) e^{-ikz} dz \right|^2. \quad (7)$$

Then, the squared magnitude of the Fourier transform can be expressed as [1]

$$|\tilde{F}(k)|^2 = \lim_{L \rightarrow \infty} \left| \int_{-L}^L f(z) e^{-ikz} dz \right|^2 = 2LW(k). \quad (8)$$

According to the Wiener–Khinchin theorem, the power spectral density $W(k)$ is equal to the Fourier transform of the autocorrelation function $R(u)$, expressed as [4]

$$R(u) = \langle f(z)f(z+u) \rangle. \quad (9)$$

Hence [1],

$$|\tilde{F}(k)|^2 = 2L \tilde{R}(k), \quad (10)$$

where $\tilde{R}(k)$ is the Fourier transform of $R(u)$.

III. RESULTS AND ANALYSIS

From the derivations, the roughness radiation loss coefficient (α_r) is obtained utilizing both Exponential and Gaussian autocorrelation functions.

Using the exponential autocorrelation function [1]:

$$R(u) = \sigma^2 e^{-\frac{|u|}{L}} \quad (11)$$

Taking the Fourier transform of $R(u)$, we get [5],

$$R(k) = \int_{-\infty}^{\infty} \sigma^2 e^{-\frac{|u|}{L}} e^{-iku} du = \frac{2\sigma^2 L}{1 + (kL)^2} \quad (12)$$

where

$$k = \beta - n_2 k_0 \cos \theta.$$

We get the radiation loss coefficient for exponential autocorrelation as stated in [1],

$$\alpha_r = \phi^2(d)(n_2^2 - n_1^2)^2 \frac{k_0^3}{4\pi n_1} \left[\int_0^\pi \frac{2\sigma^2 L d\theta}{1 + L^2(\beta - n_2 k_0 \cos \theta)^2} \right] \quad (13)$$

Similarly, for the Gaussian autocorrelation case [1]:

$$R(u) = \sigma^2 e^{-\frac{u^2}{L^2}} \quad (14)$$

Taking the Fourier transform of $R(u)$ [6], we obtain

$$R(k) = \sigma^2 L \sqrt{\pi} e^{-\frac{k^2 L^2}{4}} \quad (15)$$

where

$$k = \beta - n_2 k_0 \cos \theta.$$

Using the value of $R(k)$, we achieve the following equation for radiation loss coefficient [1],

$$\alpha_r = \phi^2(d)(n_2^2 - n_1^2)^2 \frac{k_0^3}{4\pi n_1} \times \left[\int_0^\pi \sigma^2 L \sqrt{\pi} e^{-(\beta - n_2 k_0 \cos \theta)^2 (L^2/4)} d\theta \right] \quad (16)$$

The derivation shows that the loss coefficient (α_r) depends on root mean square roughness, correlation length and waveguide modal parameters. Zadehgo and Guiana highlight that exponential autocorrelation function validates with numerical FDTD and is a good working model that focuses on simpler analytic expressions [7].

From Fig. 1, we can analyze that at very small correlation length (L), both models give minimum loss. The difference starts to notice when both models experience a peak loss with Gaussian being higher than exponential ACF. As the correlation length (L) increases more, loss for the Gaussian ACF decreases rapidly where the exponential ACF has a slower decay rate. This characteristic of exponential ACF often accurately portrays the actual roughness found in many real world waveguides and is often used for its simpler analytical solutions.

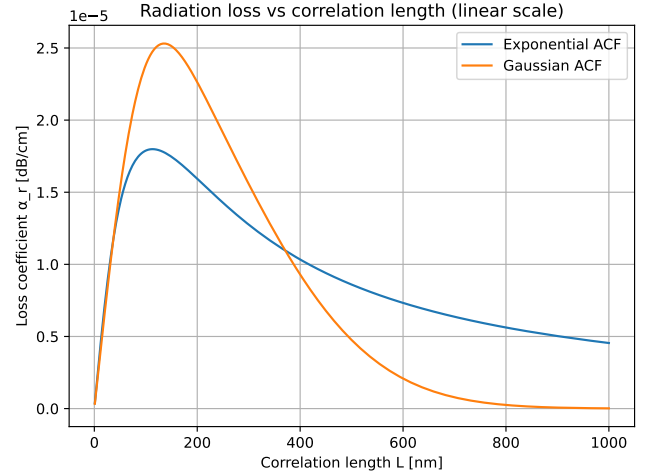


Fig. 1. Radiation loss vs correlation length for Exponential and Gaussian autocorrelation models.

IV. CONCLUSION

This work provided a detailed analytical framework for radiation loss in planar dielectric waveguides caused by side-wall roughness. Using Green's function analysis, Fourier transforms, and the Wiener-Khinchin theorem, we derived explicit formulas linking loss to roughness statistics and waveguide parameters. Comparisons of exponential and Gaussian autocorrelation models showed distinct trends: Gaussian predicts higher peak loss and faster decay, while exponential offers simpler closed-form results consistent with experiments and simulations. The exponential model's practicality makes it valuable for design and fabrication tolerance analysis. These results provide compact scaling laws for minimizing scattering and guiding ultralow-loss integrated photonic waveguide design.

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